# Ch. 26 CW Solutions

Principles of Macroeconomics

February 18, 2026

1. Argentina:

$$\begin{aligned} 1.0427^t \cdot GDP_i &= 2 \cdot GDP_i \\ 1.0427^t &= 2 \\ \log 1.0427^t &= \log 2 \\ t \log 1.0427 &= \log 2 \\ t &= \frac{\log 2}{\log 1.0427} \\ t &= 16.6 \text{ years} \end{aligned}$$

   Ghana:

$$\begin{aligned} 1.0133^t \cdot GDP_i &= 2 \cdot GDP_i \\ 1.0133^t &= 2 \\ t \log 1.0133 &= \log 2 \\ t &= \frac{\log 2}{\log 1.0133} \\ t &= 52.5 \text{ years} \end{aligned}$$

   South Korea:

$$\begin{aligned} 1.0478^t &= 2 \\ t \log 1.0478 &= \log 2 \\ t &= \frac{\log 2}{\log 1.0478} \\ t &= 14.8 \text{ years} \end{aligned}$$

2. Groland: $\$20,000(1.02^100) = \$144,892.92$
   Sloland: $\$20,000(1.015^100) = \$88,640.91$
   Groland's GDP per capita is roughly 60% higher than Sloland's in 100 years.

3. High levels of saving and investment spending are a better predictor of a future high growth rate. High levels of investment mean the capital stock will grow faster. Faster capital stock growth will lead to faster productivity growth in the short run. A high standard of living is a worse predictor because a high standard of living doesn't tell us anything about how productivity is changing over time.

4. Better education for young women will boost the level of human capital per worker in the economy. A better educated population is more likely to advance the level of technological knowledge in the economy.

5. Answers may vary for the country data.

   Interpretation 1: countries with more stable legal and political environments and countries with stronger protections on property rights create an environment that is more conducive to investing in physical capital and generating technological knowledge. As a result those countries are more productive and have a higher GDP per capita.

   Interpretation 2: Countries with a high GDP per capita have a higher average income and therefore more of an incentive to maintain stable legal and political environments and strong property rights. Their higher average income also gives them more resources with which to pursue those characteristics.

6. (a) Higher income gives people more money to invest in both preventative and curative care. It also gives them more money to spend on increasing technological knowledge related to health.

   (b) Better health outcomes boost human capital. Human capital boosts productivity. Productivity boosts income.

   (c) The relative importance determines the degree to which public policy should target health improvements or income improvements. If health is more determinative of income, then public policy should target health more. If income is more determinative of health, then public policy should target income more.

7. Peace and a coherent justice system make it easier to invest in physical and human capital because the capital isn't under threat from war or tenuous property rights. Additionally, peace and a coherent justice system create an environment in which it is easier to advance technological knowledge. If we interpret "easy" taxes as well-designed taxes, then they are a form of technological knowledge. Taxes raise revenue for public goods but generate deadweight loss and reduce production. A well-designed tax system will raise revenue for public goods while minimizing the disincentives on production.